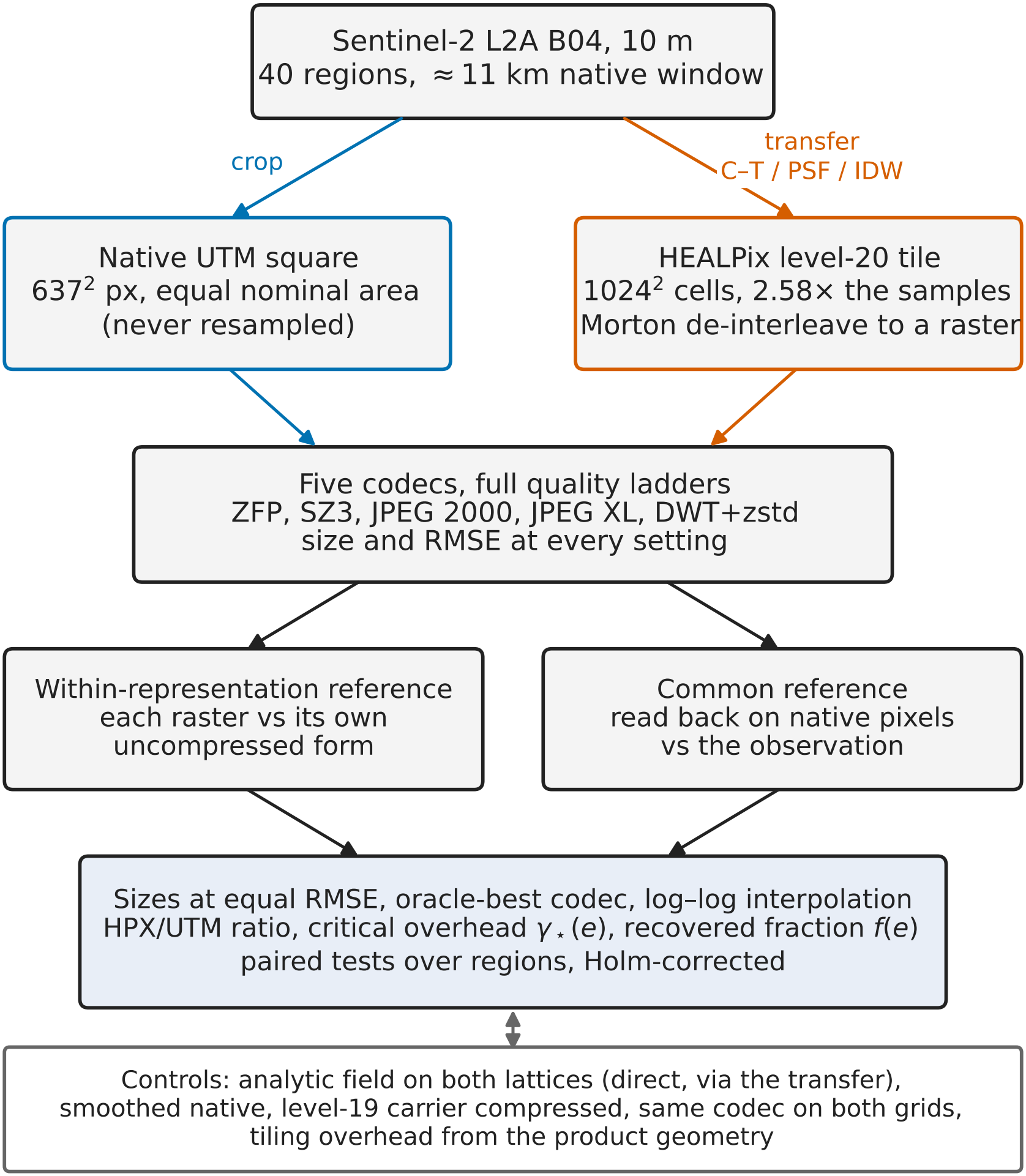
Sentinel-2 L2A B04, 10 m
40 regions, ≈ 11 km native window
crop
transfer
C–T / PSF / IDW
Native UTM square
637^2 px, equal nominal area
(never resampled)
HEALPix level-20 tile
1024^2 cells, 2.58× the samples
Morton de-interleave to a raster
Five codecs, full quality ladders
ZFP, SZ3, JPEG 2000, JPEG XL, DWT+zstd
size and RMSE at every setting
Within-representation reference
each raster vs its own
uncompressed form
Common reference
read back on native pixels
vs the observation
Sizes at equal RMSE, oracle-best codec, log–log interpolation
HPX/UTM ratio, critical overhead $\gamma_\star(e)$, recovered fraction $f(e)$
paired tests over regions, Holm-corrected
Controls: analytic field on both lattices (direct, via the transfer),
smoothed native, level-19 carrier compressed, same codec on both grids,
tiling overhead from the product geometry